

A
Mini-Project Report
On
**Smart BMS with cell balancing
system**

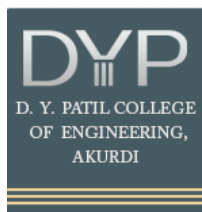
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RPROJECT GUIDE

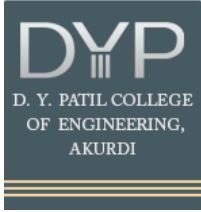
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DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION

**D.Y.PATIL COLLEGE OF ENGINEERING
AKURDI, PUNE – 411044**

2025-2026



D.Y.PATIL COLLEGE OF ENGINEERING
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DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION

CERTIFICATE

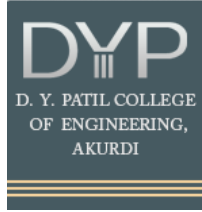
This is to certify that **Himanshu hindurao kate - T400080410** of
T.E. E&TC has completed the mini-project on

‘Smart BMS with cell balancing system’

satisfactorily under my supervision and guidance and submitted the
project report in complete fulfillment of requirement for the award of TE
(E&TC) Degree of Engineering course under the Savitribai Phule Pune
University, Pune during the academic year 2024-2025.

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CERTIFICATE

This is to certify that following students of T.E. Electronics & Telecommunication have completed the mini-project on ‘**Smart BMS with cell balancing system**’ satisfactorily under my supervision and guidance and submitted the project report in complete fulfillment of requirement for the award of Bachelors Degree of Engineering course under the Savitribai Phule Pune University, Pune during the academic year 2024-2025.

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ACKNOWLEDGEMENT

We express our sincere gratitude towards the faculty members who makes this project a successful.

We would like to express our thanks to our guide **Mrs. Meena Karad** for **her** whole hearted co-operation and valuable suggestions, technical guidance throughout the project work.

Special thanks to our H.O.D. **Dr. Rutuja Deshmukh** for her kind official support and encouragement.

We are also thankful to our mini-project coordinators **Mrs. Jaya Wakode** for her valuable guidance.

Finally, we would like to thank all staff members and faculty members of E&TC Department who helped us directly or indirectly to complete this work successfully.

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1.Introduction

In recent years, lithium-ion batteries have become one of the most widely used energy storage devices in applications such as electric vehicles, renewable energy systems, portable electronics, industrial automation, and backup power systems. Lithium-ion batteries are preferred because of their high energy density, lightweight structure, low self-discharge rate, and long life cycle. However, these batteries are highly sensitive to operating conditions such as overcharging, deep discharging, excessive current, overheating, and voltage imbalance between cells. Improper management of these parameters can reduce battery efficiency, shorten battery life, and may even lead to serious safety hazards such as thermal runaway and fire.

To overcome these problems, a Battery Management System (BMS) is used. A Battery Management System is an electronic control system that continuously monitors and manages the charging and discharging operation of battery packs. The main purpose of a BMS is to ensure battery safety, improve performance, increase efficiency, and extend battery lifespan. Modern BMS systems are widely used in electric vehicles, solar energy storage systems, laptops, drones, robotics, and other battery-powered applications.

One of the major issues in multi-cell lithium-ion battery packs is cell imbalance. During charging and discharging cycles, the voltage of individual cells may become unequal due to differences in internal resistance, temperature, aging, and manufacturing tolerances. If one cell becomes overcharged or over-discharged, the entire battery pack performance gets affected. Therefore, cell balancing becomes an essential feature in every modern BMS. Cell balancing equalizes the voltage of all cells in the battery pack and improves battery reliability, efficiency, and lifespan.

The proposed project, “Smart Battery Management System with Cell Balancing using ESP32,” is designed to develop a smart and efficient battery monitoring and protection system for a 2S lithium-ion battery pack. The system uses the ESP32 microcontroller as the main controller because of its high processing capability, built-in Wi-Fi connectivity, low power consumption, and multiple ADC channels. The system continuously monitors

important battery parameters such as total battery voltage, individual cell voltages, current, temperature, and State of Charge (SoC).

The proposed system represents a simplified prototype of modern smart BMS systems used in electric vehicles and advanced energy storage systems. Although industrial BMS systems use dedicated battery monitoring ICs and high-power MOSFET-based protection circuits, the fundamental working principles remain similar. This project demonstrates practical implementation of battery monitoring, protection, balancing, and IoT-based data monitoring in an economical and educational manner.

2.Literature Survey

2.1 Introduction to Battery Management System

A Battery Management System (BMS) is an electronic system used to monitor, control, and protect rechargeable battery packs. With the rapid growth of electric vehicles, renewable energy systems, portable electronics, robotics, and industrial automation, lithium-ion batteries have become one of the most widely used energy storage devices because of their high energy density, lightweight structure, long cycle life, and low self-discharge rate. However, lithium-ion batteries are highly sensitive to operating conditions such as overcharging, deep discharging, overheating, excessive current flow, and voltage imbalance between cells. These conditions can reduce battery performance, shorten battery lifespan, and may even lead to dangerous situations such as thermal runaway and fire. Therefore, an efficient Battery Management System is necessary to ensure battery safety, reliability, and proper operation.

2.2 Existing Battery Management Systems

Traditional Battery Management Systems mainly used analog circuits for battery protection. These systems provided basic protection against overvoltage and undervoltage conditions using comparators, transistors, and protection ICs. Although these systems were simple and economical, they lacked advanced monitoring capability, flexibility, and real-time data analysis. Modern BMS systems are microcontroller-based and provide advanced features such as voltage monitoring, current sensing, temperature monitoring, State of Charge (SoC) estimation, fault detection, and wireless communication. Industries and researchers commonly use controllers such as Arduino, STM32, and ESP32 for implementing smart BMS systems because of their high processing capability and easy interfacing with sensors and communication modules. In electric vehicles and industrial battery packs, dedicated battery monitoring ICs are used for accurate cell voltage measurement and protection. These systems also use communication protocols such as

CAN Bus for real-time communication between the battery pack and vehicle control systems.

2.3 Cell Balancing Techniques

Cell balancing is an important function of a Battery Management System used to maintain equal voltage across all cells in a battery pack. During charging and discharging cycles, cells may develop different voltage levels because of internal resistance variation, temperature changes, aging, and manufacturing differences. If the imbalance increases, some cells may become overcharged or deeply discharged, reducing battery performance and safety. Research studies show that even a voltage difference of 50–100 mV between cells can affect battery efficiency and lifespan. To overcome this problem, two major balancing techniques are used: passive balancing and active balancing.

Passive balancing is the most commonly used method in low-cost battery systems. In this method, excess energy from higher-voltage cells is dissipated through resistors using MOSFET switching circuits. It is simple, economical, and easy to implement, but some energy is lost as heat.

Active balancing transfers energy from higher-voltage cells to lower-voltage cells using capacitors or inductors. This method is more efficient and is mainly used in electric vehicles and high-capacity energy storage systems, but it increases system complexity and cost.

2.4 Battery Monitoring Parameters

Modern Battery Management Systems continuously monitor important battery parameters such as voltage, current, temperature, State of Charge (SoC), and State of Health (SoH) to ensure safe and efficient battery operation. Voltage monitoring is used to measure individual cell voltage and total battery voltage. Lithium-ion cells generally operate safely between 2.5V and 4.2V. Overcharging above 4.2V or deep discharging below 2.5V can damage the battery.

Current monitoring helps detect charging and discharging current conditions and protects the battery from overcurrent faults. Temperature monitoring is also essential because lithium-ion batteries are sensitive to overheating. Most batteries operate efficiently

between 20°C and 45°C. State of Charge (SoC) indicates the remaining battery capacity in percentage form and helps estimate battery backup time. State of Health (SoH) represents the aging condition and overall health of the battery

2.6 Limitations of Existing Systems

Although modern Battery Management Systems provide advanced protection and monitoring features, many industrial BMS systems are expensive and complex. Some low-cost systems provide only basic protection without balancing or wireless monitoring capability. Traditional analog systems also suffer from lower accuracy and limited flexibility.

2.7 Conclusion of Literature Survey

From the literature survey, it is observed that Battery Management Systems are essential for improving the safety, efficiency, and lifespan of lithium-ion batteries. Modern BMS systems monitor important parameters such as voltage, current, temperature, SoC, and SoH to protect batteries from overcharging, deep discharging, overheating, and overcurrent conditions. The survey also shows that cell balancing is important for maintaining equal voltage across battery cells and improving overall battery performance. Recent advancements in smart BMS systems include microcontroller-based monitoring and IoT technology for real-time wireless monitoring. Based on these observations, the proposed project focuses on developing a smart and low-cost Battery Management System using ESP32 with monitoring, protection, balancing, and IoT features.

3.Problem Statement with Objectives

3.1 Problem Statement

Lithium-ion batteries are widely used in modern applications, but improper battery management can lead to problems such as overcharging, deep discharging, overheating, overcurrent, and voltage imbalance between cells. These issues reduce battery efficiency, decrease battery lifespan, and may create safety risks. Many low-cost battery systems do not provide proper monitoring, balancing, and real-time protection features. Therefore, there is a need for a smart and low-cost Battery Management System that can monitor battery parameters, protect the battery from fault conditions, and improve battery performance and safety.

3.2 Objectives of the Project

The specific objectives of this mini-project are :

1. To design and develop a smart Battery Management System using ESP32 for a lithium-ion battery pack.
2. To monitor important battery parameters such as voltage, current, temperature, and State of Charge (SoC) in real time.
3. To provide protection against overvoltage, undervoltage, overcurrent, overheating, and reverse current conditions.
4. To implement cell balancing for maintaining equal voltage across battery cells.
5. To improve battery safety, efficiency, reliability, and lifespan.
6. To control charging and discharging operation using relay-based protection.

7. To implement IoT-based wireless monitoring using ESP32 and ThingSpeak cloud platform.
8. To demonstrate the basic working principle of modern smart Battery Management Systems used in electric vehicles and energy storage systems.

4.Specifications of Project

4.1 Functional Requirements:

Sr. No.	Requirement	Specification
1	Voltage Monitoring	0V – 8.4V
2	Current Monitoring	±5A (ACS712)
3	Temperature Monitoring	-55°C to 125°C
4	SoC Calculation	0% – 100%
5	Cell Balancing	Passive Balancing
6	Protection Features	OV, UV, OC, OT Protection
7	Relay Control	Automatic Load Disconnect
8	IoT Monitoring	ThingSpeak Cloud Monitoring

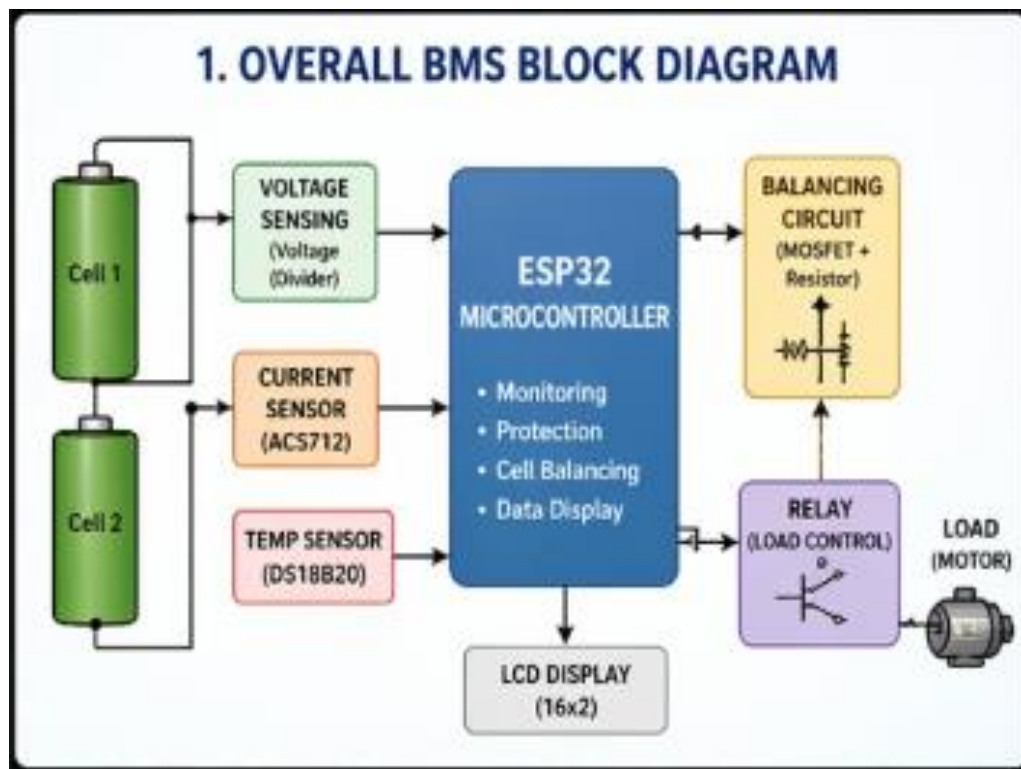
4.2 Hardware Specifications:

Sr. No.	Component	Specification
1	Microcontroller	ESP32-WROOM-32
2	Battery	2 × 3.7V Li-ion Cells
3	Current Sensor	ACS712 5A
4	Temperature Sensor	DS18B20
5	Relay Module	5V Single Channel
6	MOSFET	IRFZ44N
7	Resistors	20kΩ, 10kΩ, 220Ω
8	Power Supply	DC-DC Buck Converter
9	LED Indicator	5mm LED

4.3 Other specifications:

Sr. No.	Parameter	Value / Description	
1	Cell Balancing Method	Passive Cell Balancing	
2	Balancing Technique	MOSFET + Resistor Discharge	
3	Protection Features	Overvoltage, Undervoltage, Overcurrent, Overtemperature	
4	Relay Type	5V Single Channel Relay	
5	MOSFET Used	IRFZ44N	
6	ADC Resolution	12-bit ADC (ESP32)	
7	Voltage Divider Resistors	20k Ω and 10k Ω	
8	Operating Voltage for ESP32	5V	
9	Operating Temperature	0°C to 45°C	
10	State of Charge (SoC)	0% – 100%	
12	Programming Software	Arduino IDE	
13	Communication Protocol	Wi-Fi (TCP/IP)	

5. Block Diagram



Block diagram description

The proposed project is a Smart Battery Management System (BMS) with Cell Balancing using ESP32 designed for monitoring, protection, and efficient management of a lithium-ion battery pack. The system continuously monitors important battery parameters such as total battery voltage, individual cell voltages, current, temperature, and State of Charge (SoC) in real time.

Voltage sensing is performed using voltage divider circuits connected to the ESP32 ADC pins for measuring total battery voltage and individual cell voltages. The ACS712 current sensor is used to measure charging and discharging current, while the DS18B20 sensor monitors battery temperature. The ESP32 microcontroller processes all sensor data and performs battery monitoring and protection functions.

The system implements passive cell balancing using MOSFET and resistor circuits. When voltage imbalance occurs between battery cells, the balancing circuit discharges

the higher-voltage cell through a resistor to maintain equal voltage distribution across cells. This improves battery efficiency, safety, and lifespan.

Protection features such as overvoltage, undervoltage, overcurrent, overheating, and reverse current protection are implemented in the system. A relay is used to automatically disconnect the load during fault conditions to protect the battery and connected devices.

The project also supports IoT-based real-time monitoring using ESP32 Wi-Fi communication and ThingSpeak cloud platform. Battery parameters can be monitored remotely through graphical visualization on the cloud platform. The proposed system demonstrates the basic working principle of modern smart Battery Management Systems used in electric vehicles and energy storage applications.

6. Circuit Diagram

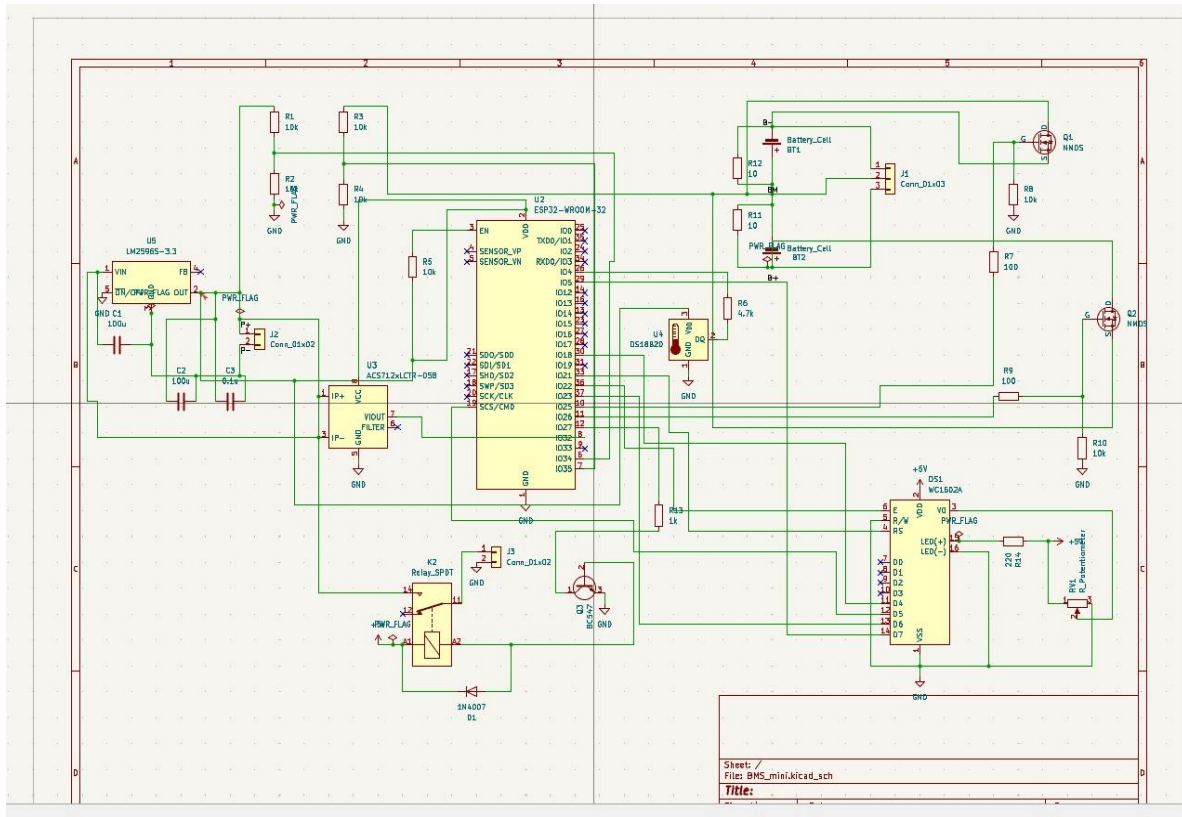


Fig. Circuit diagram of smart BMS and cell balancing circuit

Circuit Diagram Description

The circuit diagram of the proposed Smart Battery Management System (BMS) is based on the ESP32-WROOM-32 microcontroller, which controls monitoring, protection, and cell balancing operations. A 2S lithium-ion battery pack is used as the power source.

Voltage divider circuits using 20k Ω and 10k Ω resistors are used to measure total battery voltage and individual cell voltages through ESP32 ADC pins. The ACS712 current sensor measures charging and discharging current, while the DS18B20 sensor monitors battery temperature.

Passive cell balancing is implemented using IRFZ44N MOSFETs and balancing resistors. The ESP32 activates the balancing circuit when voltage imbalance occurs.

between cells. A relay module is used for load control and battery protection during fault conditions.

A DC-DC buck converter provides regulated 5V supply to the ESP32 and other components. The system also supports IoT-based monitoring using Wi-Fi and ThingSpeak cloud platform for real-time battery parameter monitoring.

7.Selection of components, calculations

Selection of Components

1. ESP32-WROOM-32 Microcontroller

ESP32 is selected as the main controller because it provides high processing speed, built-in Wi-Fi connectivity, multiple ADC channels, and low power consumption. It is used for monitoring battery parameters, controlling balancing circuits, relay operation, and IoT communication with ThingSpeak.

2. Lithium-Ion Battery (2S Configuration)

Two 3.7V lithium-ion cells are connected in series to form a 2S battery pack with a nominal voltage of 7.4V and maximum voltage of 8.4V. Lithium-ion batteries are selected because of their high energy density and rechargeable capability.

3. ACS712 Current Sensor

ACS712 current sensor is used for measuring charging and discharging current of the battery pack. The sensor provides analog voltage output proportional to current flow and supports both positive and negative current measurement.

4. DS18B20 Temperature Sensor

DS18B20 digital temperature sensor is used for battery temperature monitoring. It provides accurate temperature measurement and supports one-wire communication.

5. MOSFET (IRFZ44N)

IRFZ44N MOSFET is used in the passive cell balancing circuit. The MOSFET acts as an electronic switch to discharge excess voltage from higher-voltage cells through balancing resistors.

6. Relay Module

A 5V single-channel relay module is used for load control and protection. The relay disconnects the load automatically during fault conditions such as overvoltage, overcurrent, or overheating.

7. DC-DC Buck Converter

A buck converter is used to step down battery voltage to 5V for powering the ESP32 and other components safely.

8. Voltage Divider Circuit

Voltage divider circuits using $20\text{k}\Omega$ and $10\text{k}\Omega$ resistors are used to reduce battery voltage to a safe ADC input range for ESP32.

8. PCB Art work

The hardware implementation of the proposed Smart Battery Management System is developed using a Zero PCB (General Purpose PCB) for compact and reliable circuit assembly. After successful testing of the circuit on the breadboard, all components were mounted on the Zero PCB to obtain a stable and semi-permanent hardware setup.

The PCB layout is designed by considering proper placement of components such as ESP32, ACS712 current sensor, DS18B20 temperature sensor, relay module, MOSFET balancing circuit, voltage divider network, and connector terminals. Proper routing of power and signal connections is maintained to reduce noise and ensure accurate sensor readings.

The ESP32 module is placed at the center of the PCB for easy connection of sensors and control circuits. Voltage divider resistors are connected to ADC input pins for voltage sensing. The MOSFET and balancing resistor circuits are placed near battery terminals to simplify balancing connections. The relay module is connected for automatic load control and battery protection during fault conditions.

A DC-DC buck converter is used on the PCB to provide a regulated 5V supply to the ESP32 and other components. Proper soldering and insulation techniques are followed to ensure safe operation of the battery management system.

The developed Zero PCB-based hardware provides compact size, improved connection reliability, reduced wiring complexity, and easier handling compared to breadboard implementation. The current prototype can be further converted into a custom-designed PCB using KiCad software for industrial-level implementation and compact product development.

9. Testing Procedures

The testing of the proposed Smart Battery Management System (BMS) was carried out in multiple stages to ensure proper operation of the hardware and software modules.

Initially, the complete circuit was implemented on a breadboard for prototype testing and verification. All hardware components including ESP32, ACS712 current sensor, DS18B20 temperature sensor, relay module, voltage divider circuits, MOSFET balancing circuit, and lithium-ion battery connections were assembled on the breadboard. Proper wiring connections and sensor interfacing were verified carefully before powering the system.

After completing the hardware setup, the ESP32 program was uploaded using Arduino IDE. The Serial Monitor was used to observe real-time values of battery voltage, individual cell voltages, current, temperature, State of Charge (SoC), balancing status, and fault conditions. Different operating conditions were tested to verify voltage sensing, current measurement, temperature monitoring, relay operation, and passive cell balancing functionality. The balancing operation was tested by intentionally creating voltage differences between cells and observing the balancing status in the Serial Monitor. Protection features such as overvoltage, undervoltage, overcurrent, and overheating conditions were also verified by applying different test conditions.

After successful testing and verification on the breadboard, the circuit was shifted to a Zero PCB for permanent mounting and improved reliability. All components were soldered properly on the Zero PCB to obtain a compact and stable hardware prototype. Final testing was again performed after PCB mounting to verify correct operation of the complete Smart BMS system

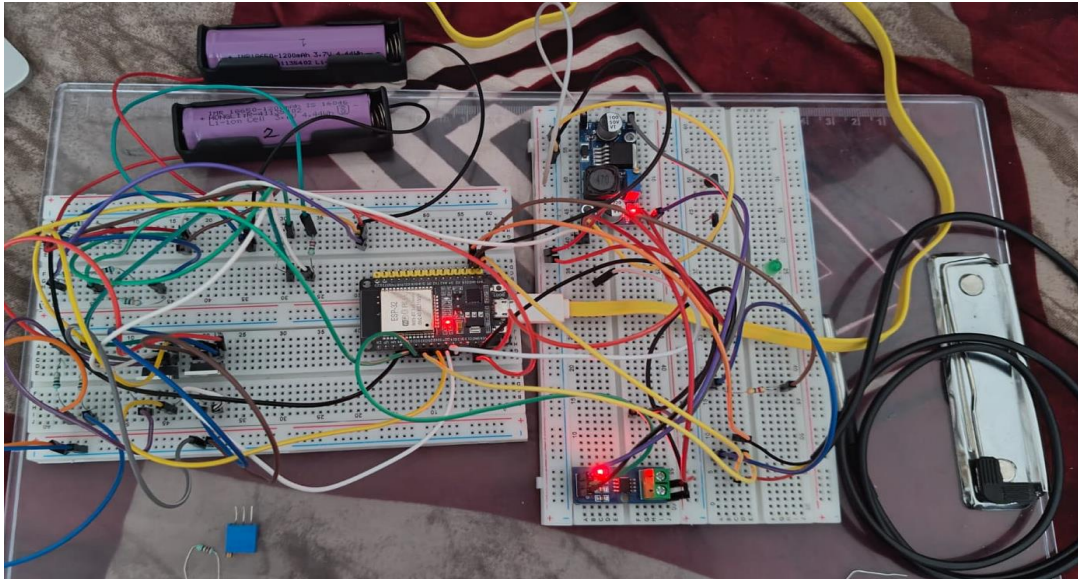


Figure 9.1: Breadboard Implementation of Smart BMS Prototype

```
ESP32 Dev Module
sketch_apr19cino
127 // ----- PROTECTION (UPDATED) -----
128 bool fault = false;
129
130 // Overvoltage
131 if (cell1 > 4.2 || cell2 > 4.2) fault = true;
132
133 // Undervoltage
134 if (cell1 < 2.0 || cell2 < 2.0) fault = true;
135
136 // Severe imbalance
137 if (abs(cell1 - cell2) > 1.0) fault = true;
138
Output Serial Monitor X
Message (Enter to send message to 'ESP32 Dev Module' on 'COM4')
C1: 2.98 C2: 3.71 Tot: 6.69 I: 0.22 T: 32.7 SoC: 53.8 Diff: 0.73 BALANCING NORMAL
C1: 2.98 C2: 3.73 Tot: 6.70 I: 0.28 T: 32.7 SoC: 52.6 Diff: 0.75 BALANCING NORMAL
C1: 2.99 C2: 3.71 Tot: 6.69 I: 0.12 T: 32.7 SoC: 51.4 Diff: 0.72 BALANCING NORMAL
C1: 2.99 C2: 3.71 Tot: 6.70 I: 0.17 T: 32.7 SoC: 50.3 Diff: 0.73 BALANCING NORMAL
C1: 2.97 C2: 3.74 Tot: 6.71 I: 0.15 T: 32.7 SoC: 49.3 Diff: 0.78 BALANCING NORMAL
C1: 2.98 C2: 3.61 Tot: 6.59 I: 0.09 T: 32.7 SoC: 48.0 Diff: 0.63 BALANCED NORMAL
C1: 2.97 C2: 3.73 Tot: 6.70 I: 0.28 T: 32.7 SoC: 47.1 Diff: 0.75 BALANCING NORMAL
C1: 2.98 C2: 3.72 Tot: 6.70 I: 0.08 T: 32.7 SoC: 46.2 Diff: 0.75 BALANCING NORMAL
C1: 2.98 C2: 3.71 Tot: 6.69 I: 0.15 T: 32.7 SoC: 45.3 Diff: 0.73 BALANCING NORMAL
C1: 3.05 C2: 3.67 Tot: 6.72 I: 0.09 T: 32.7 SoC: 44.6 Diff: 0.61 BALANCED NORMAL
C1: 2.98 C2: 3.76 Tot: 6.74 I: 0.20 T: 32.7 SoC: 43.9 Diff: 0.79 BALANCING NORMAL
C1: 3.07 C2: 3.62 Tot: 6.70 I: 0.33 T: 32.7 SoC: 43.1 Diff: 0.55 BALANCED NORMAL
C1: 3.01 C2: 3.73 Tot: 6.73 I: 0.26 T: 32.7 SoC: 42.5 Diff: 0.72 BALANCED NORMAL
Ln 137, Col 31 ESP32 Dev M
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Figure 9.2: Real-Time Testing and Monitoring of Battery Parameters

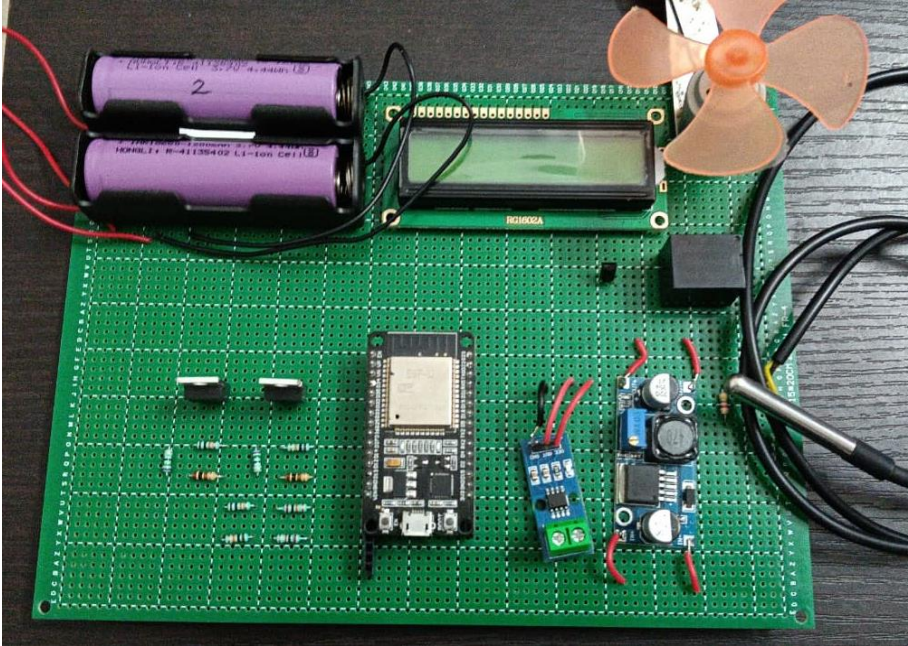


Figure 9.3: Final Hardware Prototype on Zero PCB

11. Conclusion

The Smart Battery Management System (BMS) with Cell Balancing using ESP32 was successfully developed and tested for a 2S lithium-ion battery pack. The project achieved real-time monitoring of important battery parameters including total battery voltage, individual cell voltages, current, temperature, and State of Charge (SoC). The system continuously monitored battery conditions and provided protection against overvoltage, undervoltage, overcurrent, overheating, and abnormal operating conditions.

Passive cell balancing was implemented using MOSFET and resistor circuits to maintain equal voltage distribution between battery cells. The balancing operation helped improve battery efficiency, safety, and overall battery lifespan. Relay-based load control successfully disconnected the battery during fault conditions, ensuring safe operation of the system.

The use of ESP32 provided fast processing capability along with built-in Wi-Fi communication for IoT-based monitoring. Integration with the ThingSpeak cloud platform enabled remote monitoring and visualization of battery data in real time. The project also demonstrated practical implementation of embedded systems, sensor interfacing, battery protection circuits, and IoT communication.

The hardware prototype was initially tested on a breadboard and later implemented on a Zero PCB for better stability and compactness. The project successfully demonstrated the working principle of a smart BMS used in modern battery-powered applications such as electric vehicles, solar energy storage systems, portable electronics, and backup power.

Overall, the project provided a low-cost, reliable, and efficient battery monitoring and protection solution. Future improvements may include active balancing techniques, dedicated BMS ICs, mobile application integration, custom PCB fabrication, and advanced State of Health (SoH) estimation methods for industrial-level battery management applications.

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